Souhrid Sarkar

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Personal Summary

MSc cancer researcher with expertise in 3D organoid models, CRISPR-based gene perturbation, and quantitative imaging to study tumor cell plasticity and microenvironmental adaptation. Experienced in profiling stemness-associated and stress-response pathways under therapeutic pressure. Strong interest in how cancer stem cells interact with immune components and remodel their niche to evade surveillance. Committed to rigorous, reproducible experimental research in tumor-immune biology.

Education

University of Glasgow Glasgow, UK

Master of Science (MSc) in Cancer Research and Precision Oncology 2024 - 2025

Awarded- The International Leadership Award and The India Merit Award

Drug Discovery, Clinical Trial, Core Skills in Cancer Research, Immunotherapy, Data Analysis

Master's Thesis: Understanding the Role of Autophagy in KRAS-Mutant Colorectal Cancer

Grade: A (First Class with distinction)

Amity University Kolkata, India

Bachelor of Technology in Biotechnology 2020 - 2024

Bachelors Thesis : Folate functionalized nanomedicine use for tumour targeted drug delivery in breast cancer.

Coursework -Molecular and Cell Biology, Genomics, Biochemistry, Immunology, Nanomedicine

Grade: First Division

Work Experience

Graduate Research Assistant, CRUK, Scotland Institute Apr 2025 – Sep 2025

- Ran combination drug screens on $KRAS^{G12C}$PN and $KRAS^{G12D}$PN colorectal cancer organoids, demonstrating >99.7% viability loss in G12C models with Afatinib-based regimens.
- Profiled stemness-associated and stress-response markers (e.g., EMT/stemness regulators) using confocal microscopy, flow cytometry, and Western blotting, linking phenotypic plasticity to treatment adaptation.
- Contributed to in vivo research pipelines by assisting with immunohistochemistry, immunofluorescence, and murine dissection, gaining cross-disciplinary training.
- Independently prepared qPCR workflows on treated organoids, integrating transcriptomic and phenotypic readouts for mechanistic interpretation.

Research Intern, Medical College and Hospital, Kolkata June 2023 – Dec 2023

- Processed and analysed 200+ paediatric cancer plasma samples to validate circulating protein and miRNA biomarkers for treatment-response prediction.
- Performed ELISA, HPLC, and flow cytometry assays with >95% reproducibility, enabling robust downstream statistical analysis.
- Integrated assay results with patient clinical data to identify biomarker thresholds with potential 80% predictive accuracy for disease progression.
- Performed immunophenotyping using flow cytometry to characterize immune cell populations and leukemic subtypes, ensuring accurate gating and reproducible analysis.

Research Associate, BioLim Centre for Science & Technology, Tamil Nadu July 2022 – Jan 2023

- Curated and harmonised multi-omics datasets from GEO/TCGA across 5+ breast and cervical cancer cohorts, resolving >300 metadata inconsistencies.
- Shortlisted 54 candidate biomarkers via differential expression analysis in R, increasing prioritisation efficiency for wet-lab validation by 40%.
- Mapped high-priority genes to signalling pathways in STRING/Reactome, guiding design of ELISA and IHC validation workflows.
- Delivered literature briefs synthesising biomarker function and clinical relevance, accelerating hypothesis generation for translational cancer assays.

Research Intern, Indian Institute of Technology, Bombay May 2021 – Jan 2022

- Built an RNA-seq analysis pipeline that improved pituitary adenoma subtype classification accuracy by 20%, using a 15-gene expression signature.
- Identified molecular drivers of non-functional adenomas through pathway enrichment, informing potential drug repurposing candidates.
- Simulated drug–target interactions via computational modelling, predicting three compounds with >70% binding affinity for shortlisted targets.
- Standardised bioinformatics workflows to ensure reproducibility and scalability across multiple tumour datasets.

Publications

Immunomodulatory Signaling Networks in Glioblastoma Multiforme (Peer-Reviewed, 1st Author) July 2024

- Performed an in-depth review of immunomodulatory signaling networks in glioblastoma multiforme, identifying key therapeutic targets and providing actionable insights for treatment development.
- Impact: Synthesised fragmented evidence into a framework that highlights new immunotherapy targets in glioblastoma, a highly treatment-resistant cancer. This work has been cited in ongoing reviews of glioblastoma therapy and is being used as a reference for designing immune-modulation strategies.
- Sarkar, S., & Patranabis, S. (2024). Immunomodulatory signaling networks in glioblastoma multiforme: A comprehensive review of therapeutic approaches. Human Cell. Advance online publication. https://doi.org/10.1007/s13577-024-01112-8.

Role of Extracellular Vesicles in Intercellular Communication in the Brain (Peer-Reviewed, 1st Author) Feb 2024

- Examined the role of Extracellular Vesicles (EVs) in neurodegenerative diseases, identifying their potential as biomarkers and therapeutic targets for conditions such as Alzheimer's and Parkinson's disease.
- Impact: Brought together neuroscience and nanomedicine perspectives, helping frame EVs as both biomarkers and delivery vehicles. This paper has been noted for its translational focus, supporting the development of diagnostic and therapeutic approaches in neurodegenerative research.
- Sarkar, S., & Patranabis, S. (2024). Emerging Role of Extracellular Vesicles in Intercellular Communication in the Brain: Implications for Neurodegenerative Diseases and Therapeutics. Cell Biochemistry and Biophysics. https://doi.org/10.1007/s12013-024-01221-z

Additional Projects

EGFR/HER2 Expression in Pancreatic Cancer (University of Glasgow, UK, 2025)

- Compared TCGA and GTEx transcriptomic datasets, linking EGFR/HER2 overexpression to poor prognosis and gemcitabine resistance. Delivered reproducible code pipeline and visual survival analyses, providing a rationale for dual-target therapeutic strategies. Enhanced expertise in RNA-seq analysis and clinical data integration.

SNP Harmonization Pipeline (University Hospital of Tubingen, Germany, 2025)

- Built a harmonised dataset of 885 SNPs by integrating VCF files across OpenCRAVAT, Ensembl VEP, and dbSNP. Improved downstream variant annotation reliability by 30% and produced a clean SQLite-ready database for cancer genomics applications. Gained proficiency in large-scale data cleaning, annotation, and pipeline reproducibility.

Skills

Molecular Biology & Cell Culture	Computational & Bioinformatics	Data Analysis & Compliance
CRISPR-Cas9 gene editing	DESeq2, edgeR, ggplot2	GraphPad Prism, FlowJo, Image Studio, QuPath
qPCR, Immune profiling	Python (pandas, numpy, matplotlib)	ICH-GCP, CLIA, GDP standards
Western blot, ELISA, HPLC	MySQL database management	Statistical testing (t-test, ANOVA, survival analysis)
Flow cytometry, confocal microscopy, immunofluorescence	VCF, GEO, TCGA, cBioPortal data mining, single-cell sequencing	Experimental design & QC
2D/3D mammalian cell culture	GSEA, STRING, Reactome, Transcriptomic profiling, NGS sequencing	Data visualisation & reporting
Organoid culture (colorectal)	Protein structure modelling (AlphaFold, Molecular Docking, Schrödinger)	Documentation & SOP development

Awards, Certifications & Professional Organisation

- Awarded “The India Merit Award” and “International Leadership Scholarship” by the University of Glasgow
- Awarded the “Young Researcher Award” by AlzForum
- Specialization in Genomic Data Science certification by John Hopkins University
- Certification in 3D bioprinting by NIT Rourkela
- Python Course on Multiple Languages by IIT Madras
- Associated with American Association for Cancer Research, American Society of Clinical Oncology, European Association for Cancer Research, Royal Society of Biology.

Conferences and Presentations

- Poster Presentation: “Targeting Senescence and Immune Modulation in Triple Negative Breast Cancer for Improved Prognosis and Therapy” at the European Association for Cancer Research (EACR) Conference (2024).
- Oral Presentation: “Gut Microbiota Modulation Enhances Doxorubicin-Induced Cell Death in TNBC” at the International Association for Integrated Oncology (2023).

Language Skills

- **Native languages:** Bengali, Hindi
- **English:** C1 (TOEFL)
- **German:** B1 (University Course)

References

- **Dr. Shiladitya Ghosh**
  Assistant Professor
  Amity Institute of Biotechnology
  Amity University Kolkata
  Work- mail: sghosh10@kol.amity.edu
  Email: shiladityag18@gmail.com (Currently available through email)

- **Dr. Laura Richmond**
  Co-Director, MSc Programme in Cancer Research and Precision Oncology
  Wolfson Wohl Cancer Research Centre
  University of Glasgow
  Email: laura.richmond@glasgow.ac.uk